Manchester Metropolitan University

Department of Computing and Mathematics

ASSIGNMENT COVER SHEET

Unit code and title:	6G7V0018_2526_1F | Robotics Science and Systems
Assignment set by:	Sergio Davies
Assignment ID: *(e.g., 1CWK50)*	1CWK100
Assignment weighting:	100%
Assignment title:	Robotics: Implementation and Report
Type: (Group/Individual)	Individual
Hand-in deadline:	Refer to unit Moodle space
Hand-in format and mechanism:	Portfolio (Report and Codes), individually upload from Moodle

Learning outcomes being assessed:

LO1: Understand the theoretical basis of robotics and applications.
LO2: Develop practical skills in programming robots.
LO3: Analise the state of the art in robotics models, and how this informs the design of future robot applications.
LO4: Apply a wide range of transferable skills and attributes applicable to industry and research.

Note: it is your responsibility to make sure that your work is complete and available for marking by the deadline. Make sure that you have followed the submission instructions carefully, and your work is submitted in the correct format, using the correct hand-in mechanism (e.g., Moodle upload). If submitting via Moodle, you are advised to check your work after upload, to make sure it has uploaded properly. If submitting via OneDrive, ensure that your tutors have access to the work. Do not alter your work after the deadline. You should make at least one full backup copy of your work.

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Penalties for late submission

The timeliness of submissions is strictly monitored and enforced.
All coursework has a late submission window of 7 calendar days, but any work submitted within the late window will be capped at 40%, unless you have an agreed extension. Work submitted after the 7-day late window will be capped at zero unless you have an agreed extension. See 'Assessment Mitigation' below for further information on extensions.
Please note that individual tutors are unable to grant any extensions to assessments.

Assessment Mitigation

If there is a valid reason why you are unable to submit your assessment by the deadline you may apply for Assessment Mitigation. There are two types of mitigation you can apply for via the module area on Moodle (in the 'Assessments' block on the right-hand side of the page):

- **Non-evidenced extension**: Since this is an in-class assignment, non-evidenced extensions are not allowed.
- **Evidenced extension**: requires you to provide independent evidence of a situation which has impacted you. Allows you to apply for a longer extension and is available for event-based assessment such as in-class test, presentations, interviews, etc. For event-based assessments, the normal outcome is that the assessment will be deferred to the summer reassessment period.

Further information about Assessment Mitigation is available on the dedicated Assessments page.

Plagiarism

Plagiarism is the unacknowledged representation of another person's work, or use of their ideas, as one's own. Manchester Metropolitan University takes care to detect plagiarism, employs plagiarism detection software, and imposes severe penalties, as outlined in the Student Code of Conduct and Academic Misconduct Policy. Poor referencing or submitting the wrong assignment may still be treated as plagiarism. If in doubt, seek advice from your tutor.
As part of a plagiarism check, you may be asked to attend a meeting with the Module Leader, or another member of the module delivery team, where you will be asked to explain your work (e.g. explain the code in a programming assignment). If you are called to one of these meetings, it is very important that you attend.

Use of generative AI

The use of generative AI is permitted in this assessment, so long as it is used in accordance with the instructions provided in the 'Are you allowed to use AI in assessments?' section of the AI Literacy Rise Study Pack. All submitted work must be your own original content.

If you are unable to upload your work to Moodle

If you have problems submitting your work through Moodle, you can send your work to the Assessment Management Team using the Contingency Submission Form. Assessment Management will then forward your work to the appropriate person for marking. If you use this submission method, your work must be sent **before the published deadline**, or it will be logged as a late submission. Alternatively, you can save your work into a single zip folder then upload the zip folder to your university OneDrive and submit a Word document to Moodle which includes a link to the folder. **It is your responsibility to make sure you share the OneDrive folder with the Module Leader, or it will not be possible to mark your work.**

Assessment Regulations

For further information see the Postgraduate Assessment Regulations on the Assessments and Results information pages.

Assessment Criteria:	Indicated in the attached assignment specification.
Formative Feedback:	*Weekly discussion during lecture and lab sessions will allow self and peer assessment. It also provides opportunities for formative feedback from peers and the tutor.*
Summative Feedback Format:	*General feedback will be uploaded onto Moodle as an overall description of the performance of the cohort, including what've been done well and what needs improvements.* *Personal feedback will be given regarding the portfolio quality of a specific student, including what they have done well and what can be improved.*

Assignment Description

The coursework consists of a set of **personal assignments** which cover ROS (Robot Operating System) basics, such as writing your own topics, publishers, and subscribers and finally a technical report related to robotics to gain a general understanding of robotics in AI. All personal assignments mush be completed independently. Table 1 below describes marking details for each assignment.

The assignments are provided in the lab worksheets on Moodle. You must collect all assignments and include them in your submission document, as this serves as evidence that you have engaged with the teaching materials. If you are unsure whether you have collected all the assignments, please consult your unit tutor for confirmation.

Upon completing each lecture and lab session, you should be able to complete the assignments straight away. If you encounter any challenges, review the materials or consult your tutor. Additionally, engage with your classmates to learn from each other.

While discussions with classmates are encouraged during preparation of assignments, you must complete personal assignments independently.

Assignments and Marking Schemes

The assignments are uploaded on Moodle as part of the weekly Lab worksheet.

Personal Assignment Marking Policy:

1. **Marking Scheme:**
    - You will receive marks for each step you successfully complete, regardless of whether you complete all steps.
    - Partial completion still earns marks, so ensure you submit whatever progress you've made.
    - Please refer to Table 1 for mark distributions. For personal assignment 8 (final report) the marking grid highlights the marking criteria.
2. **Submission preparation:**
    - For Personal Assignments 1 – 6, please submit each assignment embedded in a word document, or use a Jupyter Notebook, with necessary results (visuals, values, etc.) and analysis;
    - For Personal Assignments 7 and 8, your report should use clear and concise English, that is impersonal and objective. The report excludes code snippets, pictures, and tables and should be between 1,000 – 1,500 words, to describe the results when necessary;
    - You only need to embed the code related to the specific question;

- You may follow the link (https://tohtml.com/) to integrate Python code into a Microsoft word document to keep the format. You are encouraged to find your way to keep the format of the code;

- Embed all other codes in the appendix section!

- Please find in the appendix an example way of formatting your responses.

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Marking Scheme for Report – Personal Assignment 8

Criterion	Very Poor (0-19%)	Poor (20-44%)	Marginal Fail (45-49%)	Acceptable (50-59%)	Good (60-69%)	Very Good (70-85%)	Excellent (86-100%)
Structure	The report Is **incomplete** and lacks clear organisation. No logical flow between sections.	A **functional** report, but key sections are missing, making it inconsistent and difficult to follow.	The report includes most key sections, but some are underdeveloped, affecting overall coherence.	A **well-organised** report that includes all main sections, though some lack depth or clarity.	A **structured** report covering all required sections, with logical transitions and clarity.	A **well-developed** and cohesive report, demonstrating a strong structure with all sections clearly covered.	An **exceptionally structured** report, covering all sections and subsections thoroughly, with seamless flow and readability.
Comprehensiveness	Minimal report with **little to no evidence of current state-of-the-art** analysis. **No meaningful discussion** or insights into pros and cons.	**Minimal** evidence of research. **Fails to identify significant state-of-the-art solutions** or relies on non-academic sources.	**Basic** summary of the topic. Identification of pros/cons is present but **lacks technical depth** or connection to challenges.	Analysis of the "State-of-the-Art" is **mostly accurate** but tends to be **more descriptive than critical**. Identifies **standard pros and cons** without deep investigation.	**Good coverage** of the topic with relevant academic sources. **Correctly identifies** current challenges and existing solutions with their pros and cons. The "Future Works" section is **logical** and shows a clear **understanding** of the field's direction.	**Comprehensive analysis** of the chosen topic. Clearly distinguishes between **different existing solutions** and their trade-offs. The discussion of "Pros and Cons" is not just descriptive but evaluates the impact on specific applications.	**Demonstrates a profound, critical understanding of current research**. Identifies subtle nuances in "State-of-the-Art" solutions. Future work suggestions are highly original, technically feasible, and directly address identified gaps.

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Criterion	Very Poor (0-19%)	Poor (20-44%)	Marginal Fail (45-49%)	Acceptable (50-59%)	Good (60-69%)	Very Good (70-85%)	Excellent (86-100%)
Language & Clarity	The report contains **significant typos, grammar errors, or unclear writing**, making it **difficult to understand**.	Frequent typos and grammar issues affect readability. Writing is unstructured or unclear.	Mostly readable, but **some language errors remain**, affecting clarity.	Generally well-written with **few typos or grammar mistakes**. Some sections could be clearer.	**Clear and professional writing**, with minor or no typos. Uses appropriate academic language.	**Well-polished writing**, free of errors, with a **professional and academic tone**.	**Exceptional writing quality**, completely free of errors. **Highly professional** and engaging academic language.
References & Citation	**Few or no references**. Incorrect format. References are outdated or not from reputable sources.	References exist but are **poorly formatted, outdated, or from non-academic sources**.	Some references are present but **not consistently formatted** or lack relevance.	Mostly correct formatting. References are somewhat relevant but **lack depth**.	**Correctly formatted references**. Mostly up to date, with **several from credible sources**.	References are well formatted, mostly up-to-date, and include **majority from reputable conferences and journals**.	**Well-cited, up-to-date references** from **high-impact conferences and journals**, formatted perfectly.

Mark Distributions and Provisional Release Time of the Assignments

PA – Personal Assignment

Assignments	Provisional Time to Issue	No. of Questions	Marks	Total Marks	Marks Awarded
PA – 1	Week - 2	2	2	4	
			2		
PA – 2	Week - 3	1	6	6	
PA – 7	Week - 4	1	8	8	
PA – 8	Week - 4	1	30	30	
PA – 3	Week - 4	6	5	14	
			3		
			2		
			1		
			1		
			2		
PA – 4	Week - 5	4	4	8	
			1		
			2		
			1		
PA – 5	Week - 6	6	2	16	
			2		
			3		
			3		
			3		
			3		
PA – 6	Week - 7	5	2	14	
			2		
			3		
			4		
			3		

Table 1: Assignments release time and marking schemes.

Submission Guidelines

1. You need to submit all the required files related to the personal assignments in a single .zip file. You should name the zip file as ‘FirstName-SurName-MMU ID’.

2. There is no size limit for the submission.

3. If your submission exceeds 100 MB, follow these steps:

    - Upload your coursework to OneDrive.
    - Copy and paste the shareable link into a Word document.
    - Ensure that "Anyone with the link can edit" is enabled on your OneDrive upload.
    - Submit the Word document on Moodle containing the link.

Failing to set the correct permissions may result in your submission being inaccessible and ungraded.

Support

Primary Contact: Email
For support regarding learning and assessment, email is your primary point of contact with staff. To ensure a prompt response, please contact your tutor directly and include the unit and assessment title in the email subject line. In the body of the email, provide a concise outline of the specific problem and attach screenshots where appropriate to illustrate the issue. Depending on the nature of the inquiry, they may provide a direct solution via email or, if several students are experiencing similar issues, issue a group email to the class.

Secondary Contact: MS Teams
While email remains the standard method of communication, MS Teams meetings may be proposed on a case-by-case basis. These sessions are reserved for instances where more intensive support is required to help you organize, structure, or finalize your coursework. Additionally, depending on the circumstances, occasional online or in-classroom drop-in sessions may be offered.

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Appendix

Personal Assignments

*Below is an **example** way to format your answers to each question*

1. PA-1
    a. Question 1
        i. You related code snippets like below

```
def bubble_sort(arr):
     n = len(arr)
     for i in range(n):
         for j in range(0,n-i-1):
             if arr[j] >arr[j+1]:
                 arr[j],arr[j+1]=arr[j+1],arr[j]
     return arr

numbers = [64, 34, 25, 12, 22, 11, 90]
print("Sorted List:", bubble_sort(numbers))
```

        ii. Your Response/Observations to Question 1, can be visuals/texts, etc.

    b. Question 2
        i. You related code snippets like below

```
import requests
from bs4 import BeautifulSoup

url = "https://example.com" response = requests.get(url)
soup = BeautifulSoup(response.text, "html.parser")

print("Page Title:", soup.title.string)
```

        ii. Your Response/Observations to Question 2, can be visuals/texts, etc.

Follow the same style to format your response PA-1 to PA-6.
In case you have code included in your report, please follow the same style.

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Assignments

Personal Assignment 1

Write and execute a Python script named **yourfirstname_lastname.py** that calculates the **sum of numbers from 0 to 99**.

Question 1: Use relevant commands to write and execute the script.

Question 2: Document the **commands used** and explain their **purpose**.

Personal Assignment 2

After fully understanding the files in:

- ~/catkin_ws_rss/src/rss_linux_pkg/scripts
- ~/catkin_ws_rss/src/rss_linux_pkg/src

Please complete the following questions:

Question 1: Write a .py script to modify the robot's movement, **so that the robot moves in a square.**

Personal Assignment 3

Please replace 'USER' with 'yourfirstnameyourlastname'.

Question 1: Modify USER_publisher.py so that the robot moves in a straight line (e.g., along the x or y direction). Name this file USER_publisher_line.py.

Question 2: Create a launch file for USER_publisher_line.py.

Question 3: Use USER_subscriber.py to listen to the topic published by USER_publisher_line.py to confirm that the robot is indeed moving in a straight line.

Question 4: Launch the TurtleBot3 empty environment and observe its behaviour.

Question 5: Compare the behaviour of USER_publisher.py and USER_publisher_line.py, describing your observations.

Question 6: Can you stop the robot from running? *(Hint: There are two ways!)*

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Personal Assignment 4

Please replace 'USER' with 'yourfirstnameyourlastname'.
Question 1: Write a USER_pubsub.py script where you:

- Subscribe to /odom
- Publish to /cmd_vel

Question 2: Define a distance for the robot to travel. The robot should stop once it reaches the specified distance.

Question 3: Create a launch file for this script.

Question 4: Launch the TurtleBot3 empty environment and observe whether the script functions as expected.

Personal Assignment 5

1. **Create launch files** to start:

Question 1: The **publisher and subscriber**

Question 2: The **server and client**

2. **Modify the code in Section 7** so that the robot moves for **30 seconds**, following this sequence:

Question 3: First 20 seconds: The robot moves in a circle.

Question 4: Next 5 seconds: The robot stops.

Question 5: Final 5 seconds: The robot moves along the x-axis.

Question 6: Then, it stops completely.

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Personal Assignment 6

Service Message Type:

```
float64 sideLength
int32 repetitions
---
bool success
```

Question 1: **Create a turtlebot_move_square.srv message** and place it in the appropriate directory.

Question 2: **Modify CMakeLists.txt and package.xml** accordingly.

Develop a client and server that utilize the turtlebot_move_square.srv message to perform the following:
Question 3: When the server is called, the robot should move along a square, where the **side length is defined by sideLength**.

Question 4: The robot must repeat this motion according to the **value of repetitions** (e.g., if repetitions=4, the robot moves along the square **four times**).

Question 5: Once the robot completes its movement, the server should return **True**. Otherwise, it should return **False**.

Personal Assignments 7 & 8

Personal Assignments 7 & 8 together will form a technical report.

- The report (excluding pictures, tables, and references) should be **between 1,000 - 1,500 words**. Over the 1,500 words won't be penalised. Fewer than 1,000 words will be penalised.

Personal Assignment 7

- **Choose a robotics-related topic**, such as:
  1. Mobile Robot Simultaneous Localisation And Mapping (SLAM);
  2. Navigation that is related to robots;
  3. Human-robot collaboration in manufacturing/healthcare, etc.;
  4. Human-robot interaction in manufacturing/healthcare, etc.;
  5. Multi-robot coordination;
  6. Digital twin of robotic systems;

7. Quadruped robot;
8. Other: must be robotic related, please speak to the lecturer if you are not sure your topic is robotic related.

Personal Assignment 8

Based on **Personal Assignment 7**, compile a structured report following this outline:

1. **Title (You decide it)**
2. **Abstract**
3. **Background of the topic you choose, or an introduction of the topic**
4. **The challenges of the topic across academia and industry**
5. **The state-of-the-art solutions and their pros and cons**
6. **Future works to improve the solutions**
7. **Conclusion**
8. **Self-reflection of the RSS unit.**
9. **References**